Marco Vital

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Portfolio: https://marcojvital.github.io/portfolio/

Education

University of Massachusetts Amherst (Commonwealth Honors College)
- BS in Mechanical Engineering, 3.81 GPA (Cumulative), Anticipated Spring 2028

Relevant Work Experience

Atlas Actuator Team Co-op, Boston Dynamics, Waltham, MA ***July 2026- Present***
- Working on gear design for the Atlas humanoid robot. Wrote a MATLAB optimization program using KISSsoft COM Interface to optimize gear macrogeometry, balancing strength parameters, efficiency, and backlash within a given size envelope.
- Contributed to gear material analysis, factoring in impact strength, durability, cost, machinability, and heat distortion.
- Performed axial and radial tolerance stackups on actuators, reducing a critical stackup through GD&T changes.
- Ran Cpk and correlation analysis using Minitab, allowing tolerances to be relieved, improving manufacturability.
- Assisted metrology lab with gear measuring. Interpreted gear inspection reports to determine root causes of problems.

Research Assistant, Elaine Marieb Center for Nursing and Engineering Innovation, Amherst, MA ***Summer 2023-2025***
- Collaborated with graduate students and registered nurses to design and prototype engineering solutions that address challenges faced in hospital settings. Utilized SOLIDWORKS, FDM printers, Python programming, and circuit design to accomplish this. Presented my work at annual, campus-wide poster sessions.
- 2025: Created a board to determine robotic dexterity for simulated nursing tasks (peeling bandages, recapping syringes). Work will be used in a research study to determine effective robot grippers for nursing-specific bimanual imitation learning.
- 2024: Designed and manufactured a 3D scanner to generate a digital object library, enabling virtual robot training using ROS for hospital applications.
- 2023: Worked towards the development of a robot gripper to be used in hospital rooms to lighten nurses' workloads, while maintaining safety and efficiency.

Student Design Teams

UMass Amherst Formula SAE Team (Minutemen Racing) ***Fall 2024- Present***
Chassis Subteam Engineering Captain | Summer 2025 - Summer 2026
- Led a team of 6 students towards building an open-wheel racecar chassis for international collegiate events.
- Completed & maintained required technical documents while running compliance checks to ensure race eligibility.
- Designed a low-cost racing seat to maximize driver comfort, while falling within tight time constraints.

Mechanical Subteam Member, UMass Robotics Research Team ***Fall 2024- Present***
- 2025-2026: Lead of development for the legs of a quadruped robot, built for research purposes. Utilized Onshape, Ansys, and SLS printing to optimize strength and weight under a tight budget.
- 2024-2025: Worked with a cross-disciplinary team of students to develop a humanoid robot with two 7-DOF arms, a 3-DOF Orbita joint neck, and an aluminum extrusion torso. Used Onshape for CAD, prototyped using FDM printers, and parts were machined on a Tormach CNC mill.

Manufacturing Intern, UMass Advanced Digital Design and Fabrication, Amherst, MA ***Fall 2025-Spring 2026***
- Worked part-time at an additive manufacturing lab using advanced 3D printers to manufacture parts for companies, on-campus labs, and student design teams. Was responsible for machine maintenance, part cleanup, and lab safety.

Engineering Skills

Hardware: GD&T, DFM, DFA, FDM, SLS, and DMLS Printing, CNC Mill, Laser/Plasma Cutter, Soldering
Software: CATIA, SOLIDWORKS, Onshape, Ansys Mechanical, 3DEXPERIENCE, KISSsoft, Minitab
Programming: Java, MATLAB, Python